

Architect’s Brief

Digital Document Approval System

1. Architecture Purpose

This brief prepares the Digital Document Approval System for system architecture modeling. It identifies the likely system boundary, external participants, internal responsibilities, integrations, data ownership concerns, workflows, and quality drivers.

This brief is not a product requirements document and does not define implementation design or code structure.

2. System Boundary

The subject system is the **Digital Document Approval System**.

The system owns document submission, metadata capture, workflow routing, approval task management, document version tracking, signature coordination, notification coordination, search and status visibility, retention status tracking, audit history, and integration status tracking.

The system does not own enterprise-wide records policy, legal interpretation of signatures, identity proofing outside configured integrations, external signature ceremony execution, external document repository governance, or collaborative document editing.

3. External Participants

Participant	Role in the Architecture
Document Submitter	Uploads documents, enters metadata, starts workflows, responds to requested changes, and tracks status.
Approver	Reviews documents and records approval, rejection, or change request decisions.
Reviewer	Provides comments or feedback without final approval authority.
Workflow Administrator	Configures workflow templates, routing rules, escalation rules, roles, and retention settings.
Records Manager	Reviews retention status, archival eligibility, legal holds, and records-related tasks.
External Signer	Signs documents through an external signature workflow.

Participant	Role in the Architecture
Compliance Auditor	Reviews approval evidence, audit history, signature status, and retention records.
Identity Provider	Authenticates users and provides role or group information.
E-Signature Provider	Executes signature ceremony and returns signature status and artifacts.
Notification Provider	Sends approval, reminder, escalation, signature, and completion notifications.
Document Storage Provider	Stores document binaries and archives where configured.
Enterprise Document Repository	Optional external repository for publishing approved documents or importing source documents.

4. Candidate Internal System Responsibilities

Responsibility Area	Description
Document Intake	Handles upload, metadata capture, validation, and workflow submission.
Document Version Management	Maintains document versions and ensures approval decisions are tied to the correct version.
Metadata Management	Stores and validates document metadata used for routing, search, access, and retention.
Workflow Configuration	Manages workflow templates, stages, routing rules, escalation rules, and signature requirements.
Workflow Orchestration	Runs workflow instances, advances stages, handles rework loops, and completes workflows.
Approval Task Management	Manages review, approval, rejection, change request, delegation, reassignment, and task completion.
Signature Coordination	Coordinates signature requests, status updates, completed artifacts, and failure handling with an external provider.
Notification Management	Sends assignment, reminder, escalation, signature, rejection, change request, and completion notifications.

Responsibility Area	Description
Document Search and Status	Provides document search, filtering, status visibility, and task lists.
Access Control	Enforces document, workflow, task, administrative, retention, and audit access rules.
Retention and Archival Management	Applies retention categories, archival rules, legal hold, and retention status tracking.
Audit Logging	Records document lifecycle, workflow, approval, signature, access, retention, and integration events.
Integration Monitoring	Tracks failed notification, storage, identity, repository, and e-signature integrations.
Reporting and Evidence	Provides operational reporting and audit evidence views.

5. Candidate Data Stores

Data Store	Architectural Purpose
Document Metadata Store	Stores document identity, metadata, status, owner, type, confidentiality level, and retention category.
Document Version Store	Stores version records, version numbers, upload history, approval eligibility, and current version pointer.
Document Binary Store	Stores document files directly or references an external storage provider.
Workflow Template Store	Stores reusable workflow definitions, routing rules, stages, and escalation settings.
Workflow Instance Store	Stores running and completed workflow instances, current stage, state, and related document version.
Approval Task Store	Stores assigned tasks, due dates, decisions, comments, delegation, and completion state.
Signature Request Store	Stores signature requests, signer details, provider references, signature status, and completion evidence.
Notification Store	Stores notification requests, recipients, delivery status, failures, and templates.

Data Store	Architectural Purpose
Access Policy Store	Stores permissions, role mappings, document-level access rules, and confidentiality rules.
Retention Store	Stores retention categories, archival status, legal holds, retention deadlines, and retention actions.
Search Index Store	Stores derived searchable document metadata and status for query performance.
Reporting Store	Stores derived metrics for workflow performance, overdue tasks, retention status, and audit evidence.
Integration Status Store	Stores external request status, retry state, failures, and correlation references.
Audit Log Store	Stores significant user, system, document, approval, signature, access, and retention events.

6. Candidate External Interfaces

Interface	Provider	Consumer	Notes
Document Submission Interface	Digital Document Approval System	Submitter-facing experience	Supports upload, metadata entry, workflow submission, and withdrawal.
Approval Task Interface	Digital Document Approval System	Approver and reviewer experience	Supports task retrieval, comments, approval, rejection, and change requests.
Workflow Administration Interface	Digital Document Approval System	Administrator experience	Supports workflow template, routing, escalation, signature, and retention configuration.
Document Search Interface	Digital Document Approval System	User-facing and auditor experiences	Supports search, filters, status, version history, and audit evidence

Interface	Provider	Consumer	Notes
Retention Management Interface	Digital Document Approval System	Records manager experience	discovery. Supports retention categories, archival, legal hold, and retention review.
E-Signature Request Interface	E-Signature Provider	Digital Document Approval System	Creates signature requests and retrieves signature status or signed document references.
E-Signature Event Interface	Digital Document Approval System	E-Signature Provider	Receives signature completion, decline, failure, and expiration events.
Notification Delivery Interface	Notification Provider	Digital Document Approval System	Sends approval, reminder, escalation, signature, and completion notifications.
Identity and Role Interface	Identity Provider	Digital Document Approval System	Authenticates users and resolves groups or roles.
Document Storage Interface	Document Storage Provider	Digital Document Approval System	Stores and retrieves document binaries or archives where external storage is configured.
Repository Publishing Interface	Enterprise Document Repository	Digital Document Approval System	Optional future interface for publishing approved documents or importing source documents.

7. Primary Workflows for Architecture Modeling

Workflow	Summary
Submit Document	User uploads document, enters metadata, selects workflow, and starts approval.

Workflow	Summary
Validate Metadata	System validates required metadata and resolves routing inputs.
Route Approval	System assigns tasks according to workflow configuration.
Review Document	Reviewer or approver views document and adds feedback or decision.
Request Changes	Approver returns document to submitter for revision.
Upload Revised Version	Submitter uploads a new version and required approval steps restart.
Complete Approval	Required approvers approve the current version and workflow advances or completes.
Request Signature	System sends document package and signer details to e-signature provider.
Process Signature Event	System receives signature completion, decline, failure, or expiration and updates workflow.
Escalate Overdue Task	System notifies alternate users or managers when task deadline is missed.
Archive Document	Records Manager or policy-driven action archives approved or expired document.
Apply Legal Hold	Records Manager applies legal hold and system prevents conflicting retention actions.
Retrieve Audit Evidence	Auditor searches document and reviews approval, signature, access, and retention evidence.

8. Key State Models

Document State

Candidate states:

1. Draft.
2. Submitted.
3. In review.
4. Changes requested.
5. Rejected.
6. Approved.

7. Awaiting signature.
8. Signed.
9. Canceled.
10. Archived.
11. Expired.
12. On legal hold.

Document Version State

Candidate states:

1. Uploaded.
2. Current draft.
3. Submitted for approval.
4. Superseded.
5. Approved.
6. Rejected.
7. Archived.
8. Locked.

Approval Task State

Candidate states:

1. Assigned.
2. Viewed.
3. Commented.
4. Approved.
5. Rejected.
6. Changes requested.
7. Delegated.
8. Reassigned.
9. Overdue.
10. Canceled.

Workflow Instance State

Candidate states:

1. Not started.
2. Running.
3. Waiting for approval.
4. Waiting for changes.
5. Waiting for signature.
6. Completed.

7. Rejected.
8. Canceled.
9. Failed.
10. Suspended.

Signature Request State

Candidate states:

1. Pending creation.
2. Sent.
3. Viewed.
4. Signed.
5. Declined.
6. Failed.
7. Expired.
8. Completed.
9. Canceled.

Retention State

Candidate states:

1. Not classified.
2. Active.
3. Eligible for archival.
4. Archived.
5. Eligible for deletion.
6. On legal hold.
7. Retention overridden.
8. Deleted.

9. Key Quality Drivers

Quality Driver	Architectural Implication
Auditability	Document, approval, signature, access, workflow, and retention events require durable event history.
Integrity	Approved versions must be immutable or protected from unauthorized modification.
Confidentiality	Document contents and metadata require access control by role, ownership, confidentiality, and workflow participation.
Version Correctness	Approval decisions must be bound to exact

Quality Driver	Architectural Implication
Workflow Reliability	document versions. Workflow state must remain consistent when notifications, signatures, storage, or identity integrations fail.
Compliance Support	The system must provide clear evidence for approval, signature, retention, and access reviews.
Searchability	Users and auditors need reliable discovery by metadata, status, task, version, and retention attributes.
Retention Safety	Archival, deletion, and legal hold actions must be controlled and auditable.
External Integration Resilience	Signature, notification, storage, identity, and repository interactions require failure visibility and retry paths.
Usability	Approvers and submitters need clear task state and next action visibility.

10. Architecture Risks

Risk	Architectural Concern
Approval-version mismatch	Users may approve one version while another version becomes current unless version binding is strict.
Signature workflow ambiguity	External signature events may arrive late, duplicated, failed, or out of order.
Access control complexity	Permissions may depend on department, ownership, role, workflow stage, confidentiality, and legal hold.
Workflow rule sprawl	Conditional routing and escalation rules may become difficult to reason about and test.
Document storage failure	Binary storage failures could leave metadata and workflow state inconsistent.
Notification dependence	Missed notifications may cause stalled workflows if task state is not independently visible.
Audit log sensitivity	Audit records may contain sensitive metadata and require access restrictions.
Retention conflict	Legal hold, archival eligibility, deletion

Risk	Architectural Concern
	policy, and active workflow state may conflict.
External signer identity	External signers may not have normal internal accounts, requiring separate access patterns.
Search index staleness	Search results may not reflect current workflow or retention state unless index updates are reliable.

11. Open Architecture Questions

1. Should document binaries be stored internally, externally, or both?
2. Should metadata and document content have separate access policies?
3. Which document types require external signature?
4. Should signature occur before final internal approval, after final approval, or as one workflow stage among others?
5. How should duplicated or out-of-order signature provider events be handled?
6. Which document changes require full workflow restart?
7. Can a document have multiple active approval workflows?
8. Can approvers delegate tasks, and who can approve delegation?
9. Should external signers access the system directly or only through the signature provider?
10. What audit events require access-download tracking?
11. Should legal hold block archive, deletion, editing, or all three?
12. How should retention policy changes affect documents already approved?
13. What search fields are mandatory versus optional?
14. Which notification failures require user-visible action?
15. What operational tooling is required to restart, cancel, repair, or manually advance failed workflows?

12. Recommended Modeling Scope

For the initial architecture model, focus on:

1. Document submission and metadata validation.
2. Document version management.
3. Sequential approval workflow.
4. Approval task handling.
5. Change request and revised version workflow.
6. Notification delivery.
7. Audit logging.

8. Basic search and status visibility.
9. Role-based access control.
10. Basic retention category tracking.

Defer advanced modeling for:

1. Parallel approval routing.
2. Conditional workflow rules.
3. External e-signature integration.
4. Legal hold management.
5. External document storage provider integration.
6. Enterprise repository publishing.
7. Advanced compliance evidence export.
8. Policy-based access control.